# Year 9 Science Test 1 Core paper

Biology (40%): variation, adaptation, natural selection, cells · Chemistry (20%): states of matter, atoms · Physics (40%)

**DRAFT** Not ready to print for students: Part 3: Physics (16 marks) has not been added yet.

| Name | Class | Date | Score / 40 |
|---|---|---|---|
| | | | |

**Instructions · Time: 40 minutes · Total: 40 marks**

- You have 40 minutes. There are three parts: Biology (16 marks), Chemistry (8 marks) and Physics (16 marks). Answer all the questions.
- Spend about one minute on each mark: 16 minutes on Biology, 8 minutes on Chemistry and 16 minutes on Physics.
- The last question in each part is harder (Stretch and Challenge). Answer the other questions in that part first.
- Write your answers in the spaces on this paper. The marks for each part of a question are shown in brackets, for example [2].

**Key words 关键词**

| | | | | | |
|---|---|---|---|---|---|
| **abiotic factor** | 非生物因素 | **biotic factor** | 生物因素 | **continuous variation** | 连续变异 |
| **genetic information** | 遗传信息 | **nucleus** | 细胞核 | **fertilisation** | 受精 |
| **zygote** | 受精卵 | **nature reserve** | 自然保护区 | **breeding programme** | 繁育计划 |
| **gene bank** | 基因库 | **natural selection** | 自然选择 | **gene** | 基因 |
| **generation** | 世代 | **cell wall** | 细胞壁 | **chloroplast** | 叶绿体 |
| **vacuole** | 液泡 | **magnification** | 放大倍数 | **eyepiece lens** | 目镜 |
| **objective lens** | 物镜 | **particle** | 粒子 | **solid** | 固体 |
| **liquid** | 液体 | **gas** | 气体 | **proton** | 质子 |
| **neutron** | 中子 | **electron** | 电子 | **atomic number** | 原子序数 |
| **mass number** | 质量数 | **electron configuration** | 电子排布 | **periodic table** | 元素周期表 |
| **group** | 族 | | | | |

**Command words 指令词:** Calculate 计算 · Complete 完成（填写） · Draw 画出 · Explain 解释 · Tick 打勾 · Use 使用（运用） · Write 写出

## Part 1: Biology

**16 marks · about 16 minutes**

*Questions 1 to 5 are Core. Question 6 is Stretch and Challenge.*

### 1 Environmental variation

(3 marks)

**(a)** Put one tick in each row to show whether the factor is abiotic (non-living) or biotic (living). **[2]**

| Factor | Abiotic | Biotic |
|---|---|---|
| temperature | | |
| predation (being hunted) | | |
| amount of light | | |
| infectious disease | | |

**(b)** Which one of these shows continuous variation? Tick one box. **[1]**

- ☐ blood group
- ☐ height
- ☐ tongue rolling (can or cannot)
- ☐ natural eye colour

## 2 Inherited variation

(2 marks)

**(a)** Complete the sentences using words from the box. One word is not needed. **[2]**

| **nucleus** | **fertilisation** | **environment** |
|---|---|---|

Genetic information is stored in the _____ of a cell.

A sperm cell and an egg cell join together at _____ to make a zygote.

## 3 Adaptation and extinction

(2 marks)

**(a)** Draw one line from each way of protecting a species to how it works. **[2]**

| | |
|---|---|
| **nature reserve** | endangered animals are bred in captivity to increase their numbers |
| **breeding programme** | seeds, gametes or cells are stored at a low temperature |
| **gene bank** | an area where the habitat and the species living in it are protected |

## 4 Natural selection

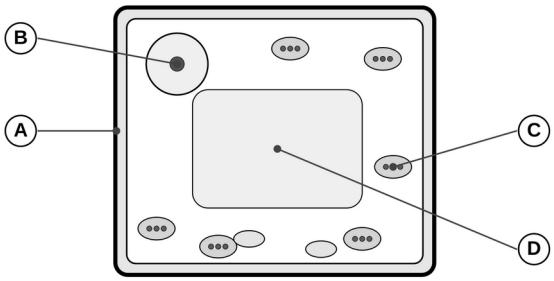
(2 marks)

**(a)** In the 1800s, black peppered moths became more common near Manchester. The sentences below explain why, but they are in the wrong order. Write 1, 2, 3 and 4 in the boxes to put them in the correct order. **[2]**

| Order | Sentence |
|---|---|
| | Birds spotted and ate more pale moths than black moths. |
| | A few moths had genes that made them black. |
| | Each generation had more black moths than the one before. |
| | Soot from factories made the tree trunks black. |

## 5 Cells and the microscope (3 marks)

**(a)** The diagram shows a plant cell. Write the name of each labelled part. **[2]**




*A plant cell seen through a microscope (not to scale)*

**A** _____

**B** _____

**C** _____

**D** _____

**(b)** A microscope has a x10 eyepiece lens and a x4 objective lens. Calculate the total magnification. **[1]**

Total magnification = x _____

## 6 Natural selection in the Salton Sea STRETCH AND CHALLENGE (4 marks)

**(a)** The Salton Sea in California has become saltier and saltier since 1905. Most fish species died out, but tilapia still live there and can cope with very salty water. Explain how natural selection made the tilapia able to live in very salty water. **[4]**

# Part 2: Chemistry

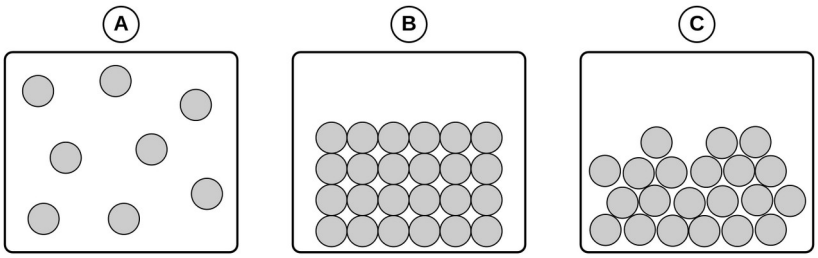
**8 marks · about 8 minutes**

*Questions 7 and 8 are Core. Question 9 is Stretch and Challenge.*

## 7 States of matter

(2 marks)

**(a)** The diagrams show the particles in a solid, a liquid and a gas. Write solid, liquid or gas for each diagram. **[2]**




**A** _____

**B** _____

**C** _____

## 8 Atoms and the periodic table

(4 marks)

**(a)** A sodium atom has atomic number 11 and mass number 23. **[2]**

**(i)** How many protons does the atom have?

protons = _____

**(ii)** How many neutrons does the atom have?

neutrons = _____

**(b)** A sodium atom has 11 electrons. Complete its electron configuration. **[1]**

2, 8, _____

**(c)** Use your answer to part (b). Which group of the periodic table is sodium in? Tick one box. **[1]**

- ☐ group 1
- ☐ group 2
- ☐ group 7
- ☐ group 0

## 9 Elements in the same group STRETCH AND CHALLENGE

(2 marks)

**(a)** Fluorine (atomic number 9) and chlorine (atomic number 17) are both in group 7 of the periodic table. **[2]**

**(i)** Write the electron configuration of each atom.

fluorine _____ chlorine _____

**(ii)** Explain why fluorine and chlorine react in similar ways.

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## Part 3: Physics

**16 marks · about 16 minutes**

*Questions from 10 onwards. The last question will be Stretch and Challenge.*

*Physics questions to be added: 16 marks, about 16 minutes. Planned as 12 marks of Core questions and one 4-mark Stretch and Challenge question, matching Part 1. The topics taught in physics are needed before these can be written.*

**END OF TEST**